



AI-Powered Invoice Processing and Expense Classification System

P R E S E N T A T I O N 2 0 2 6



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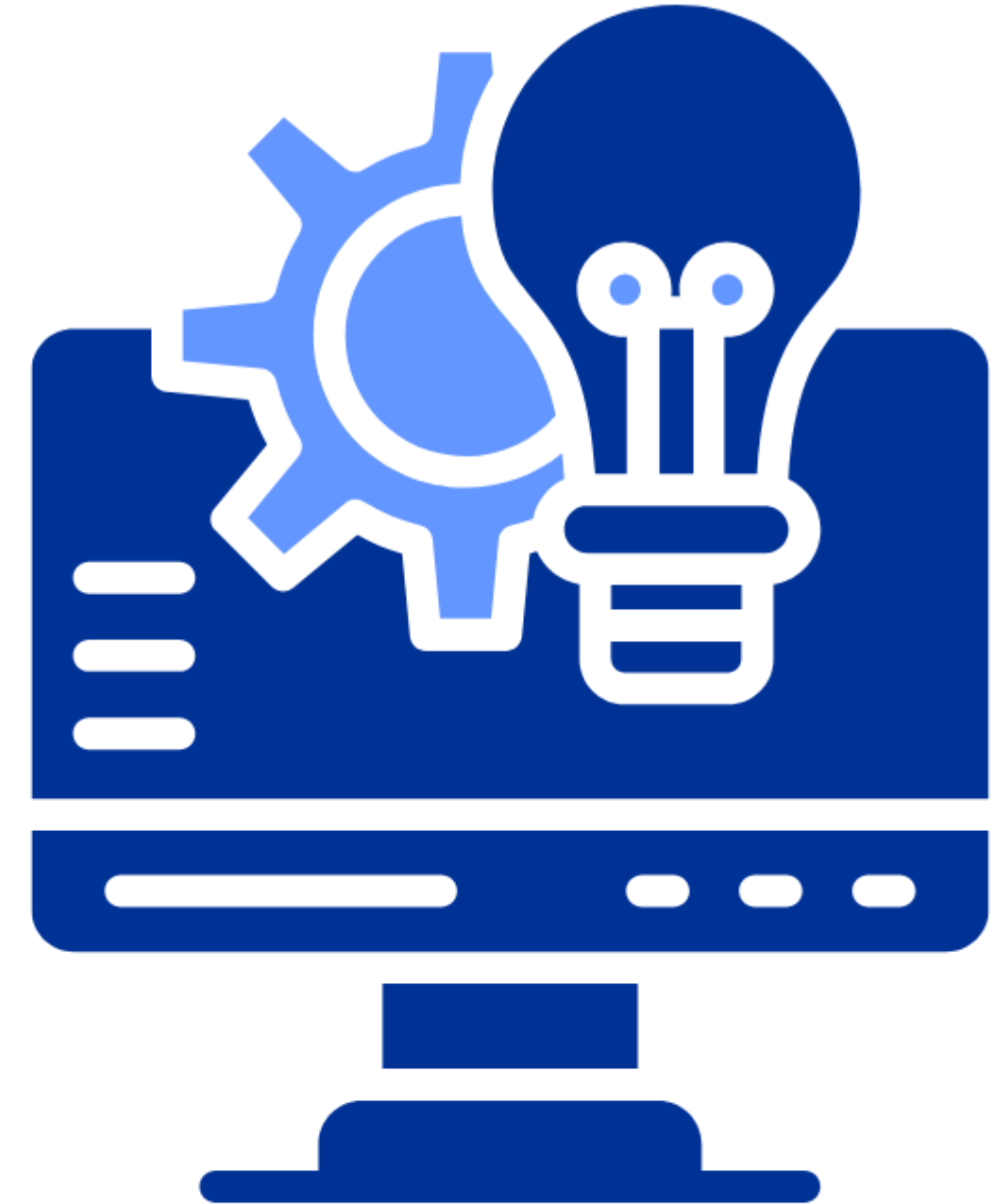
Guide

Mr. Umang Tyagi



OUTLINE

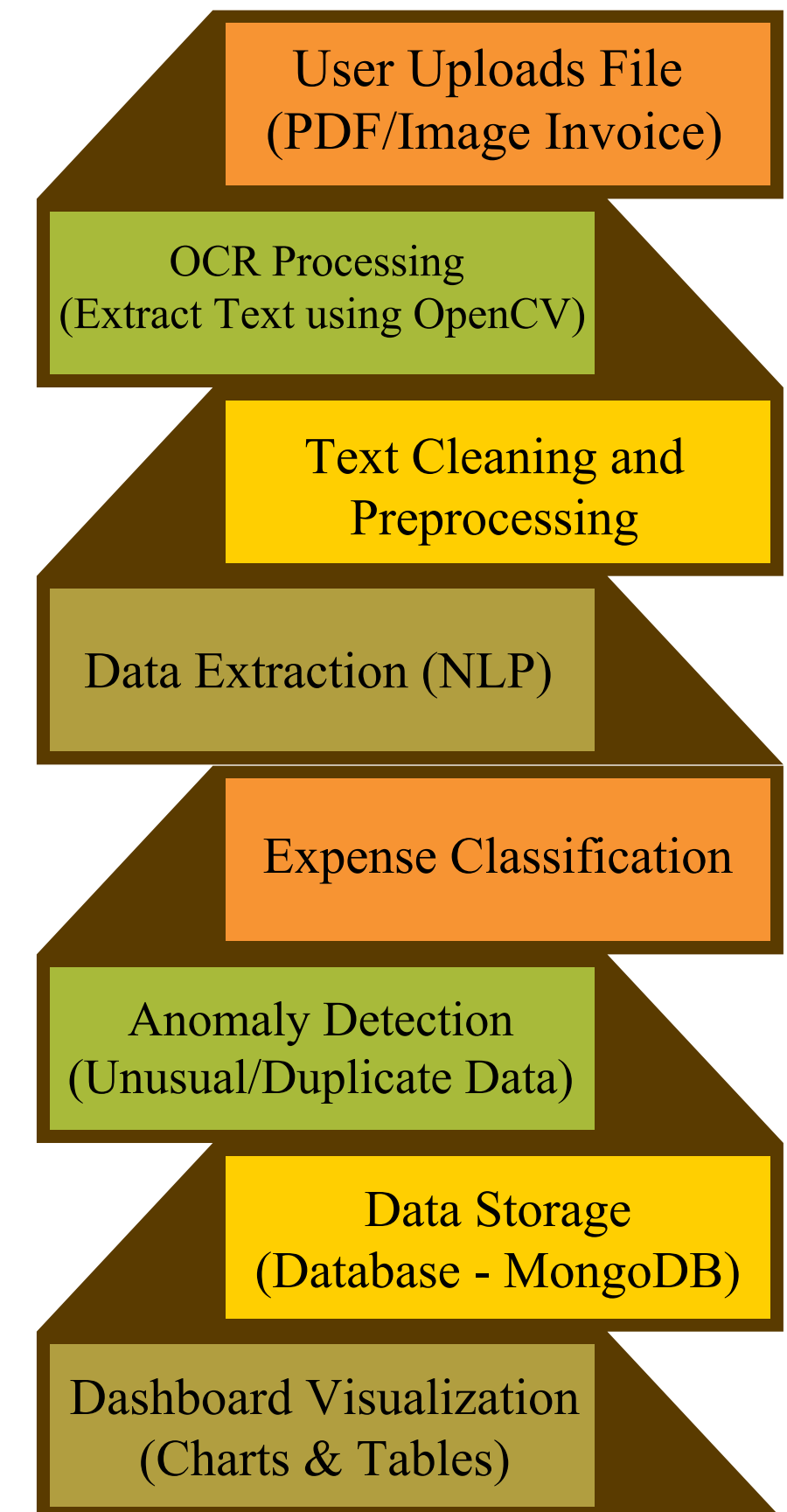
- Abstract of the Project
- Problem Statement
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Abstract

This project is an AI-based Invoice Processing System that helps in analyzing and categorizing invoices using OCR (Optical Character Recognition) and NLP (Natural Language Processing) as shown in the flowchart.

The processed data is then displayed on a dashboard with charts and tables, making it easy to understand and manage expenses. This system reduces manual work, saves time, and improves accuracy.



Problem Statement

In many organizations and daily use, invoice processing is still done manually, which takes a lot of time and effort.

People have to read each invoice, extract important details, and categorize expenses by themselves.

This process is slow, repetitive, and prone to human errors. It becomes even more difficult when there are large numbers of invoices or when invoices are in different formats like images or PDFs.

There is a need for a smart system that can automatically extract, analyze, and organize invoice data to make the process faster, easier, and more accurate.



Proposed Solution

To solve the problem of manual invoice processing, we developed an AI-based Invoice Processing System.

The system allows users to upload invoice files such as images or PDFs. It uses Optical Character Recognition technology to automatically extract text from the invoice.

After extracting the text, the system identifies important details like vendor name, invoice number, date, and total amount. It also categorizes the expenses into different categories using intelligent logic.

All the processed data is displayed on a dashboard with charts and tables, making it easy to analyze and manage invoices.

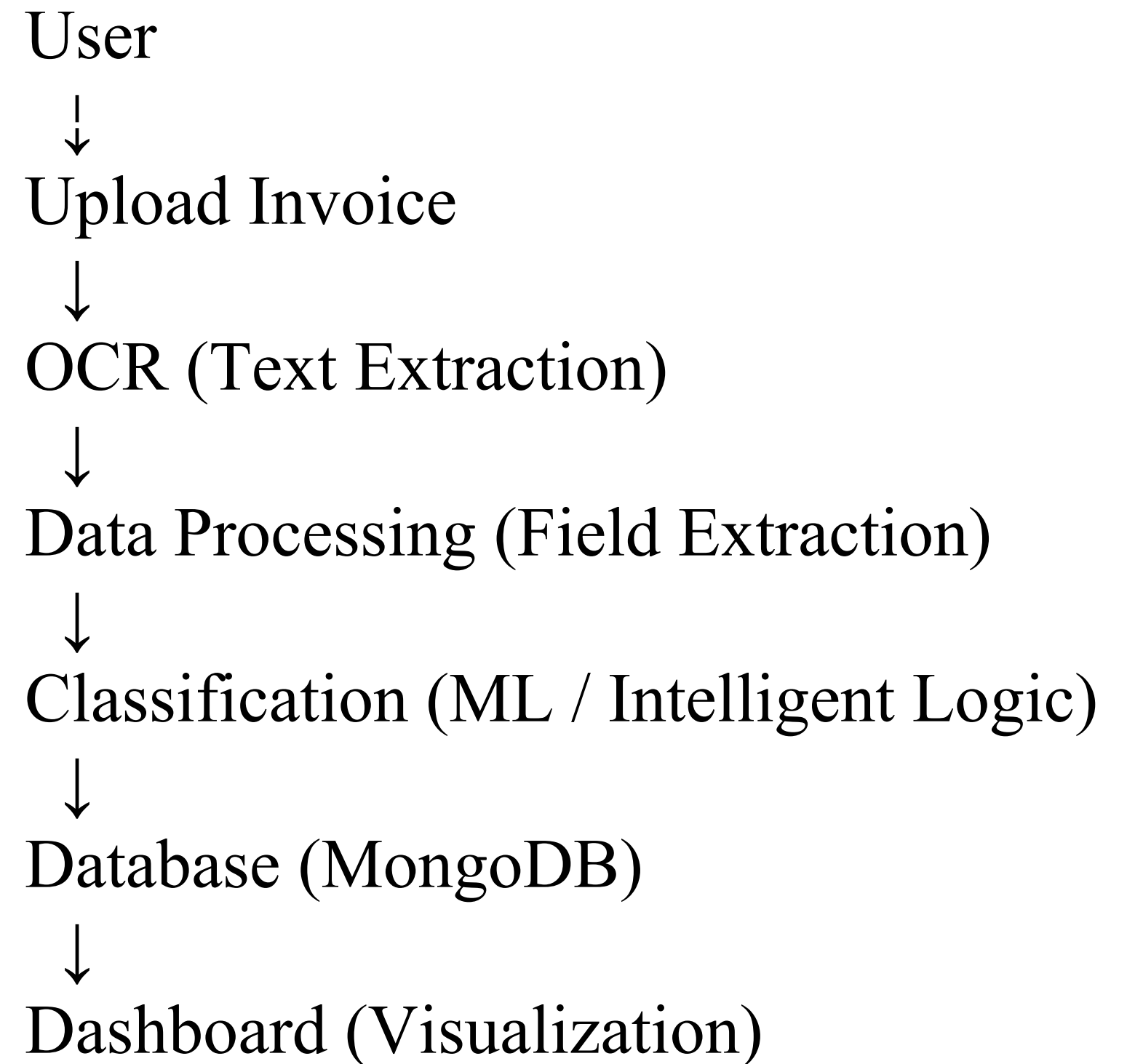
This solution saves time, reduces human errors, and improves efficiency.



System Architecture

System Architecture Overview

The system processes invoices through multiple stages starting from user upload, followed by OCR-based text extraction, data processing, intelligent classification, and storage in the database. The final results are displayed through an interactive dashboard.



Front-End and Back-End Architecture

Front-End (React)

User-friendly interface

Upload invoice files

Display extracted data

Show dashboard with charts

View invoice details



Back-End (Node.js)

Handles API requests

Manages file uploads

Processes invoice data

Communicates with database

Sends data to frontend

OCR and Intelligent Processing

1. OCR Layer

- Extracts text from invoice
- Supports images and PDFs
- Converts image → readable text

2. Data Processing

- Identifies fields
- Vendor name
- Invoice date
- Total amount

3. Classification

Categorizes expenses

Uses intelligent logic / ML model

Assigns category (Travel, Food, etc.)

Generates confidence score

The system processes extracted text and applies intelligent classification to automatically categorize invoice data, which is represented in the forms of charts, tables and bar graphs.

Data Storage and Visualization

1. Database (MongoDB)

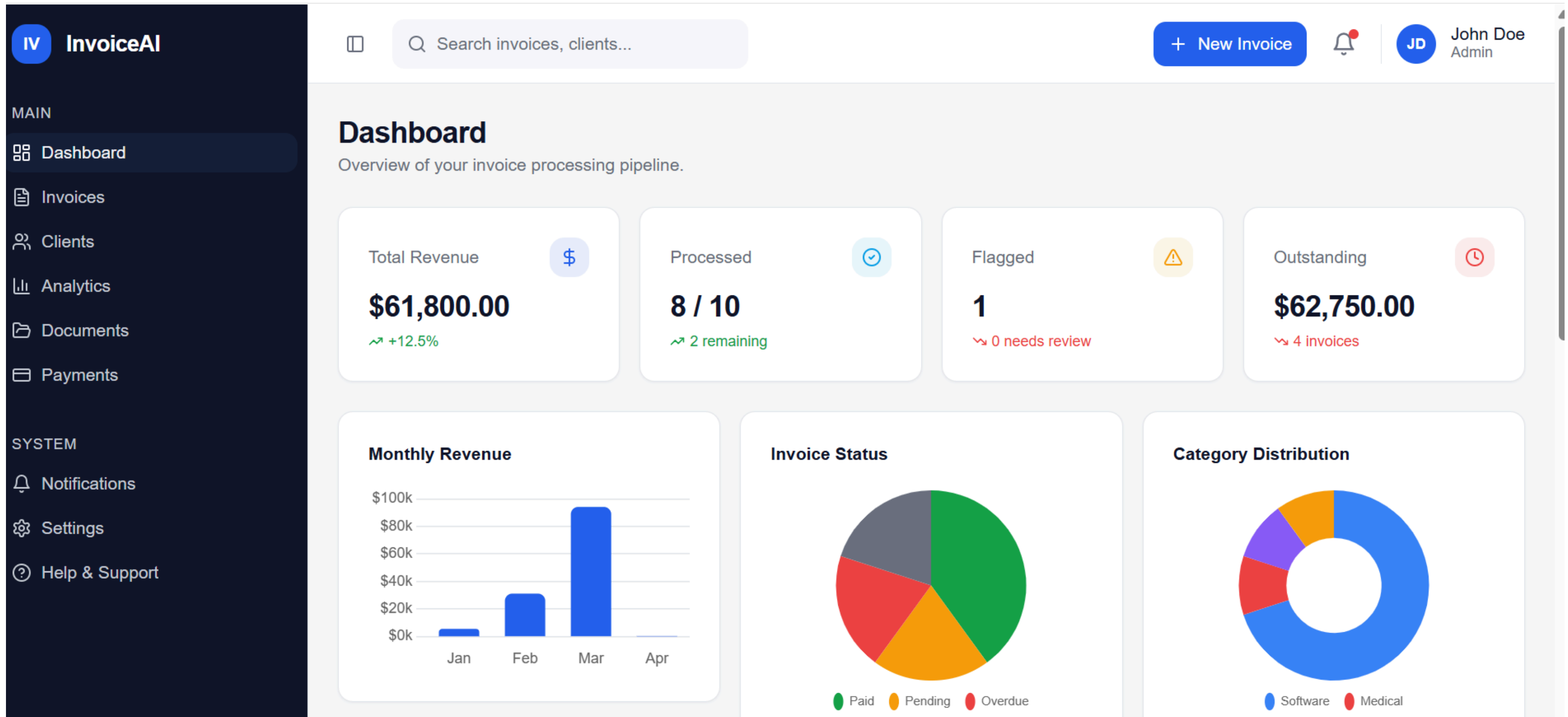
- Stores all invoice data
- Saves extracted fields
- Vendor, date, amount, category
- Stores confidence score and status

2. Dashboard (Visualization)

- Displays processed invoice data
- Shows charts and analytics
- Category-wise distribution
- Total revenue and invoice status
- Easy data understanding

The processed data is stored in the database and displayed through an interactive dashboard for better analysis and decision-making.

Live Demo of Project - Screenshots



IV InvoiceAI

MAIN

Dashboard

Invoices

Clients

Analytics

Documents

Payments

SYSTEM

Notifications

Settings

Help & Support

Log out



Search invoices, clients...

+ New Invoice



John Doe
Admin

Recent Invoices

[View all](#)

Invoice	Client	Amount	Category	Confidence	Status	Flags	Action
INV-2024-001	Acme Corp	\$12,500.00	Software	90%	Paid	—	
INV-2024-002	TechStart Inc	\$8,750.00	Software	85%	Pending	—	
INV-2024-003	Global Finance Ltd	\$24,000.00	Software	90%	Paid	—	
INV-2024-004	RetailMax	\$5,200.00	Software	75%	Overdue	—	
INV-2024-005	HealthPlus	\$15,800.00	Medical	75%	Paid	—	
INV-2024-006	EduLearn	\$3,400.00	Software	60%	Draft	—	

Conclusion

- Developed an automated invoice processing system
- Used OCR for extracting text from invoices
- Extracted important fields like vendor, date, and amount
- Applied intelligent classification for expense categorization
- Displayed results through an interactive dashboard
- Reduced manual effort and improved efficiency

The system successfully simplifies invoice management by automating data extraction, classification, and visualization.



Future Scope

- Integrate advanced machine learning models for better accuracy.
- Improve OCR to handle complex and low-quality invoices.
- Add multi-language support.
- Deploy the system on cloud for real-time access.
- Integrate with accounting and billing software.

The system can be enhanced into a fully intelligent and scalable financial management solution.



Thank you

